Sheikh Shakib Hossain

Robotics, Embedded Systems & Autonomous Systems
Researcher & Robotics Educator

Department of Computer Science and Engineering, United International University
Dhaka, Bangladesh

+880 1920 589868 | sheikhshakibpro@gmail.com | GitHub | LinkedIn

Professional Summary

Robotics, Embedded Systems & Autonomous Systems engineer, researcher, and robotics educator with hands-on experience spanning embedded firmware, sensor interfacing, autonomous robotic platforms, IoT systems, computer vision, and AI/ML. Experienced in designing and implementing basic hardware circuits, developing firmware and custom sensor libraries for NVIDIA embedded platforms, ROS2-based autonomous navigation, multi-sensor fusion, path planning, robotic manipulation, and real-time IoT monitoring. Co-Team Lead of the UIU Mars Rover Team with international competition experience and approximately three years of hands-on technical training and guidance of robotics team members and interns.

Education

2022 – 2026 **Bachelor of Science in Computer Science and Engineering**, *Department of Computer Science and Engineering, United International University*, Dhaka, Bangladesh.
CGPA: 3.82 out of 4.00

2019 – 2020 **Higher Secondary Certificate**, *Govt. Science College*, Dhaka, Bangladesh.
GPA: 5.00 out of 5.00

Technical Expertise

Robotics & Autonomous Systems: ROS2, Autonomous Navigation, Sensor Fusion, Path Planning, Computer Vision, Robotic Manipulation, Robot System Integration

Embedded Systems & Hardware: Hardware Circuit Design, Embedded Firmware, Microcontrollers, Embedded Linux, Sensor Interfacing, GPS, IMU, Hardware–Software Integration

IoT & Connected Systems: IoT Architecture, Connected Devices, Real-Time Monitoring, Device Communication, Distributed IoT Systems, Sensor-Based Monitoring

Professional Experience

2026 **IoT & Junior Full-Stack Engineer**, *Edusoft Consultants Ltd.*.
Developing hardware automation systems and connected-device applications, with emphasis on device-to-application data flow, real-time monitoring, databases, and web-based system integration.

2026 **Co-Team Lead**, *UIU Mars Rover Team*.
Led software development and system integration for the rover, contributing to autonomous navigation, sensor fusion, computer vision, path planning, robotic manipulation, hardware-software integration, testing, and field deployment. Designed and implemented embedded firmware, and wrote custom GPS and IMU libraries for NVIDIA embedded platforms where direct software support was unavailable. Trained and guided new team members and interns in robotics software development and autonomous systems.

2025 **Team Member – ARC**, *UIU Mars Rover Team*.

Took a break from University Rover Challenge (URC) to improve academics, while continuing robotics activities through participation in the Anatolian Rover Challenge (ARC). Contributed to Rover Satellite communication protocol development for the competition.

2024 **Software Team Lead**, *UIU Mars Rover Team*.

Led the software team in developing and testing rover systems, including autonomous navigation, sensor integration, computer vision, and robotic software. Contributed to system integration, debugging, and competition preparation.

2023 **Team Member**, *UIU Mars Rover Team*.

Joined the UIU Mars Rover Team and contributed to hands-on rover development, software implementation, system testing, and robotics engineering activities.

2025 – 2026 **Robotics Instructor**, *Space Innovation Camp*.

Teaching robotics, embedded systems, prototyping, debugging, and project-based engineering through hands-on technical activities and student projects.

Research Experience

2026 **Research Assistant**, *Center for Smart Infrastructure Resilience and Sustainability (CSIRS), UIU*.

Working on the project *Revise and Update the National Strategy for Water and Sanitation in Hard-to-Reach Areas of Bangladesh 2012*. Contributing to research activities and hardware-software oriented project development.

2025 – 2026 **Student Research Assistant**, *IAR-Funded IoT Campus Project*.

Worked on IoT Campus, a smart campus platform for distributed IoT data collection, real-time monitoring, data pipelines, and analytics-ready integration. Contributed to sensor-based system development, hardware planning, and hardware-software integration.

Achievements

International Rover Competition Rankings.

URC 2026: **3rd Worldwide**; URC 2024: **5th Worldwide**; URC 2023: **1st in Asia**; ARC 2025 & 2023: **3rd Worldwide**.

Autonomous Navigation Awards.

Best Autonomous Navigation System, URC 2026; **Best Autonomous Drive Award**, ARC 2025.

Research Interests

Robotics, Autonomous Systems, Embedded Systems, Internet of Things, Sensor Fusion, Computer Vision, Machine Learning, Smart Infrastructure, Intelligent Transportation Systems

Languages

English Professional Working Proficiency

Bengali Native

Referees

Md. Abid Hossain
Lecturer, Department of CSE
United International University
Email: abid@cse.uiu.ac.bd

Dr. Hasan Sarwar
Professor & Dean, SoSE
United International University
Email: hsarwar@uiu.ac.bd

Dr. Md. Shohrab Hossain
Professor, Department of CSE
United International University
Email: shohrab@cse.uiu.ac.bd